□ Excellent [10]
□ Very Good [9]
□ Good [8]
□ Satisfactory [7]
□ Bad [6]

Bangladesh University of Textiles

Department of Fabric Engineering

FE 352-0723: Fabric Manufacturing - II (Practical)

Dept: _____ Date: _____

Name: ID:

Exp No: **09** Name of the Experiment: **Study Determination of Stitch Length (SL), Wales per Inch (WPI), Courses per Inch (CPI) from Given Sample and Production Calculation**

Sign:	Key words: Stitch length, significance of stitch length, SL, WPI and CPI calculation of single jersey structure, Production calculation.

1. What is stitch length (SL)? How is the SL of a weft knitted fabric measured?

2. Why is stitch length important? What are the affected areas due to change in stitch length?

3. Determination of Stitch Length (SL) from Given Sample.

- Two points are marked (as such that there are 100 loops in between them) at the edge of the fabric from where a full course is unraveled.
- Then a full course is withdrawn from the fabric.
- The course is held at one end by the observer while a certain weight is hanged at the other end to straighten it.
- Then the marked two points are identified on the unraveled course and the value of the distance between these two points is measured with the help of a vertically held ruler.
- The procedure is repeated 4 more times.
- The average of the values is calculated.
- Then the value is divided by 100 (no. of loops) to calculate the stitch length. Generally, it is expressed in millimeter.

No. of Observations	No. of loops	Straightened Yarn Length (cm)	Average of Straightened Yarn Length (cm)	Stitch Length (cm)	Stitch Length (mm)
1	100				
2					
3					

4. Determination of Courses per Inch (CPI) from the Given Sample.

- The courses are unraveled from the cut sample one after another.
- When any of the courses touches the top side of the square, then the counting for CPI should be started.
- The courses will be unraveled one by one until the other side of the square is touched by any of the unraveled courses. Then the counting should be stopped.
- The obtained value should be put in the table.
- The same procedure is repeated for other squares.
- Then the average values are calculated to obtain the CPI of the sample.

No. of Observations	Courses per Inch (CPI)	Average CPI
1		
2		
3		

5. Determination of Wales per Inch (WPI) from the Given Sample.

- A square with an area of 1 $inch^2$ (1inch x 1inch) is drawn on the face side of the sample using ballpoint pen with two sides of it being parallel to the wales. Another 2 similar squares are drawn in such a way so that none of the wales or courses of a square is repeated in other squares.
- The number of wales exist between the two sides of the square is counted for all three squares.
- The average is calculated to obtain the WPI of the sample.

No. of Observations	Wales per Inch (WPI)	Average WPI
1		
2		
3		

6.